ly229 at georgetown.edu & (215) 688-6404

Lu Yu, Ph.D.

Interests

Quant Research & Agentic AI on Economics

Education

Ph.D., Georgetown University, Economics, (Mentors: Prof. Toshihiko & Prof. James Angel) 2019 ~ 2025
B.S., University of Pittsburgh, Mathematics & Economics (**Honor, 4.0/4.0**) 2015 ~ 2018

Skills

Certificate: Nvidia Deep Learning, Investment Banking by J.P. Morgan

Program, Software Skills and Database:

- **Tools**: Codex, Harbor, LangGraph, Pytorch, R, STATA, SQL, AWS(EC2, S3), LaTeX, Excel, MATLAB
- **Derivatives Pricing & Validation**: Equity derivatives, barrier options, multi-asset structures, Monte Carlo simulation, Black–Scholes, local and stochastic volatility, PDE methods, Greeks and sensitivity analysis, convergence testing, backtesting, P&L analysis
- **Econometrics & Risk**: Econometrics, time-series analysis, market microstructure, market trends, macroeconomic analysis, financial risk analysis, forecasting, portfolio analysis

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Work Experiences

Morgan Stanley Associate, Equity Valuation — Baltimore, MD, Aug 2025 ~ Present

- Develop AI-AGENT workflow for Risk Management. (Co-Pilot & CODEX CLI)
- Price equity derivatives, including barrier options and multi-asset structures.
- Validate pricing models under Black–Scholes and stochastic-volatility frameworks through Greek and sensitivity analysis, convergence testing, backtesting, and P&L analysis.
- Conduct risk modeling across AML, crypto, financial crime, and market risk domains, including market manipulation, insider trading, and spoofing.
- Ranked as a top performer, selected for **Super Tier 1** projects.
- Demonstrated strong execution delivering time-critical risk reports across global teams.

Georgetown University, Ph.D — Washington, DC 2020 ~ 2026

- **Instructor: Department of Finance**
  OpenFinTech: An open source project to build customized DeepAgent AI Workflow
  Design an AI Assistant for FINRA CRCP Grading (deepagent powered)
  Conducted literature reviews, prepared talking points, maintained an accurate research database.
  Performed quantitative data analysis, applying statistical methods to draw actionable conclusions.
- **Research Assistant: McDonough School of Business**
  Deep Learning on financial data analysis. Modeling Currency Volatility by using RNN(LSTM).
  Analyzed exchange-rate drivers, periodic behavior;
  Applied Monte Carlo Simulation to evaluate financial risk and model uncertainty.
- **Teaching Assistant: Department of Economics**
  Supported courses: corporate finance, financial econometrics, valuation & risk management.
  Key topics: Regression analysis, capital structure, free cash flow valuation & derivatives markets.
  Trained students to examine, manipulate, and present data in Excel & JMP.

Publications & Patents & Presentations

- PhD Thesis: ESSAYS ON INDEX OWNERSHIP AND MARKET MICROSTRUCTURE **Lu Yu**, Thesis (2026)
- From Clerks to Agentic-AI: How will Technology Change Labor Market in Finance? **Lu Yu**, Xiang Li, arXiv (2026)